

Notes: Barber & Odean (RFS, 2008)

All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Why this setting is useful	4
2.2	Large discount brokerage dataset	4
2.3	Small discount brokerage dataset	5
2.4	Large retail brokerage dataset	5
2.5	Institutional money manager dataset	5
2.6	The three attention proxies	6
2.7	The buy-sell imbalance outcome	6
2.8	Additional partitions and robustness objects	7
3	Models and empirical specifications (all equations + interpretation)	7
3.1	Economic mechanism and hypotheses	7
3.2	Volume-sort design	8
3.3	Return-sort design	8
3.4	News-sort design	8
3.5	Inference and interpretation	8
3.6	Interaction specification: volume, return, and news together	9
3.7	Size and announcement robustness	9
3.8	Short-sale-constraint test	9
3.9	Appendix model	10
4	Identification and empirical design (what makes the evidence credible)	10
4.1	What identifies the core evidence	10
4.2	Why “net buying” is stronger than “high trading”	11
4.3	Why the lagged-return design matters	11
4.4	Imperfect attention proxies and measurement error	11
4.5	Why the institutional comparison matters	11
4.6	Why the large-cap evidence matters	12
4.7	Why announcement-day robustness matters	12

4.8	Short-sale constraints as an alternative explanation	12
4.9	What the paper can and cannot distinguish	12
4.10	Relation to the literature	13
5	Tables and figures	13
5.1	Table 1 and Figures 1a and 2a: abnormal trading volume	13
5.2	Table 2 and Figures 1b and 2b: previous-day returns	14
5.3	Table 3: news sorts	15
5.4	Figure 3: volume, return, and news together	15
5.5	Table 4 and Figure 4: size and announcement robustness	17
5.6	Table 5: stocks already owned	17
6	Short summary	18
7	Conclusion	19
7.1	Core conclusion (what the paper establishes)	19
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** Are individual investors net buyers of stocks that grab their attention, and is this tendency stronger for individuals than for professional investors?
- **Setting.** The paper studies actual U.S. stock trades from three brokerage datasets for individual investors and one dataset for institutional money managers. The evidence spans the 1990s and combines brokerage records with CRSP price and volume data and Dow Jones news coverage.
- **Core challenge.** High attention days are also high trading-activity days. So a raw increase in trading says little. The paper therefore focuses on *buy-sell imbalance*: are investors more likely to buy rather than sell when a stock becomes salient? It must also distinguish true attention effects from strategy, short-sale constraints, small-stock effects, and firm-specific news.
- **Main conceptual contribution.** The paper shifts the peer of analysis from *how much investors trade* to *how attention changes the composition of trading*. The key idea is that attention primarily affects the *choice set* for buying. Preferences matter after attention narrows the list of stocks under consideration.
- **Main hypotheses.**
 - **Buying versus selling asymmetry:** attention should matter more for *buying* because investors choose from thousands of possible stocks, while selling usually involves only the few stocks already held.
 - **Individual versus professional asymmetry:** professionals have more time, better screens, and often can short. They should therefore be less prone to attention-driven buying.
- **Main empirical design.** The authors sort stocks by three attention proxies:
 - same-day abnormal trading volume,
 - previous-day one-day return,
 - same-day news coverage.

For each investor group and each partition, they compute daily buy-sell imbalances and then average them over time.

- **Main baseline result.** Individual investors are net buyers of attention-grabbing stocks. For the large discount brokerage, buy-sell imbalance by number of trades rises from **-18.15%** in the lowest abnormal-volume decile to **+29.50%** in the highest-volume vingtile.
- **Key return result.** Individuals buy both yesterday's big losers and yesterday's big winners. For the large discount brokerage, number imbalance is **+29.38%** for the worst-return vingtile, falls near zero in the middle return deciles, and rises back to **+23.98%** for the best-return vingtile.
- **Key news result.** Investors are more likely to buy stocks that appear in the news. For the large discount brokerage, number imbalance is **9.35%** on news days versus **2.70%** on no-news days. For the large retail brokerage, it is **16.17%** on news days versus **-1.84%** on no-news days.

- **Key contrast with professionals.** Institutional investors do *not* display the same attention-driven pattern. Value managers are aggressive net buyers on **low-volume** days, and momentum managers buy winners and sell losers in a way that looks like strategy rather than generalized attention.
- **Main credibility result.** The patterns remain when the paper:
 - looks separately at small, medium, and large capitalization stocks,
 - conditions on earnings and dividend announcement windows,
 - and focuses only on stocks already owned by the investor to test the short-sale-constraint alternative.
- **Economic takeaway.** Individual investors appear to buy what catches their eye. Attention changes who buys which stocks and can help propagate temporary demand pressure and sentiment, even if it does not improve returns.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper observes *actual trades*, not survey answers or stated intentions.
- The logic of the paper is strongest when the choice set is large. Common-stock purchases fit this well because investors can potentially choose from thousands of stocks.
- The contrast between buying and selling is plausible in the data. In the large discount-brokerage sample, the mean household held a monthly average of **4.3** stocks worth **\$47,334**, while the median household held **2.61** stocks worth **\$16,210**.
- Short selling is rare for these households: only **0.29%** of positions are short positions, or **0.78%** by value.

Interpretation. The empirical environment fits the theory: buying involves a massive search problem, while selling usually means choosing among a handful of already owned stocks.

2.2 Large discount brokerage dataset

- The first dataset covers **78,000 households** from **January 1991 through December 1996** for positions and through **November 1996** for trades.
- Of these households, **66,465** held common stocks at least once.
- Roughly **60%** of account market value was held in common stocks.
- In December 1996, these households held more than **\$4.5 billion** in common stock.
- During the sample period they made **1,082,107 purchases** and **887,594 sales** of common stocks.

- Mean trade size was **\$11,205** for purchases and **\$13,707** for sales; aggregate purchase and sale values were both about **\$12.1–12.2 billion**.

Interpretation. This is the paper’s flagship individual-investor sample and the one that shows the cleanest attention patterns.

2.3 Small discount brokerage dataset

- The second dataset comes from a smaller discount brokerage that marketed high-quality execution.
- It covers **January 1996 through 15 June 1999**.
- Professional accounts are excluded.
- The sample contains **14,667 accounts**, with **214,273 purchases** and **198,541 sales**.
- Mean trade value is large: **\$55,077** for purchases and **\$55,999** for sales.

Interpretation. These investors are likely more sophisticated than the large discount-brokerage customers, which makes them a useful comparison group within retail investors.

2.4 Large retail brokerage dataset

- The third dataset covers **30 months ending in June 1999**.
- The sample is restricted to **665,533 nondiscretionary accounts** with at least one common-stock trade.
- These investors execute **3,974,998 purchases** and **3,219,299 sales** of common stocks.
- Mean trade value is **\$15,209** for purchases and **\$21,169** for sales.

Interpretation. This is a very large retail sample and helps show that the attention patterns are not peculiar to one broker.

2.5 Institutional money manager dataset

- The institutional data come from the **Plexus Group**, a consulting firm that tracks trades for institutional clients.
- The sample spans **January 1993 through March 1996**.
- It includes **43** money managers, classified as:
 - **18 momentum managers**,
 - **11 value managers**,
 - **14 diversified managers**.

- The sample records both executed purchases and sales and the date and time managers decided to trade.
- Trade sizes are huge relative to retail investors. For example, momentum managers' mean purchase value is **\$886,346**.

Interpretation. These institutional data are explicitly more *illustrative than representative*, but they are still very useful as a comparison group because the paper's hypothesis is fundamentally about the difference between individuals and professionals.

2.6 The three attention proxies

The paper does not observe attention directly, so it uses three indirect measures.

(i) **Abnormal trading volume.** For stock i on day t ,

$$AV_{it} = \frac{V_{it}}{\bar{V}_{it}}, \quad \bar{V}_{it} = \frac{1}{252} \sum_{d=t-252}^{t-1} V_{id}. \quad (1)$$

Stocks are sorted each day into deciles by AV_{it} , and the highest decile is split into two vingtiles.

(ii) **Previous-day return.** Stocks are sorted each day on the basis of the *previous day's* one-day return. The top and bottom deciles are each split into two vingtiles.

(iii) **Same-day news.** Using the Dow Jones News Service for **1994–1999**, stocks are partitioned into those with and without a news story that day. On an average day, **91%** of CRSP firms have no news.

Interpretation.

- Abnormal volume is the broadest salience proxy.
- Extreme returns capture the idea that both big gains and big losses become salient.
- News is the most literal attention measure, but also the noisiest because the paper cannot observe the *intensity* or salience of each story.

2.7 The buy-sell imbalance outcome

For investor group g , partition p , and day t , buy-sell imbalance is

$$BSI_t^p = \frac{\sum_{i=1}^{n_t^p} NB_{it} - \sum_{i=1}^{n_t^p} NS_{it}}{\sum_{i=1}^{n_t^p} NB_{it} + \sum_{i=1}^{n_t^p} NS_{it}}, \quad (2)$$

where NB_{it} is the number of purchases and NS_{it} is the number of sales of stock i on day t within partition p .

The paper also computes **value imbalances** by replacing trade counts with trade values.

Additional details:

- Only **executed** trades are counted.
- If a day-partition cell has fewer than **five** trades, it is dropped from the time-series average.
- Standard errors are calculated with a **Newey–West** correction for serial dependence.

Interpretation. The paper is not asking whether high-attention stocks trade more; that is obvious. It asks whether the trading that occurs tilts toward *buying* for some investor groups.

2.8 Additional partitions and robustness objects

The paper creates several extra objects to sharpen interpretation.

- A **3-by-3-by-news** partition: abnormal volume is binned into bottom 30%, middle 40%, and top 30%; returns are binned the same way; the partition is then split by news versus no news.
- **Market-capitalization groups:** small, medium, and large, based on NYSE monthly market-equity breakpoints (below 30th percentile, 30th–70th percentile, above 70th percentile).
- **Announcement windows:** earnings announcement days and dividend announcement days.
- **Already-owned stocks:** imbalances calculated only for purchases and sales of stocks already in the investor’s account.
- **Exchange-traded options split:** used to assess whether the findings are really about short-sale constraints.

Interpretation. These objects are not afterthoughts; they are central to the paper’s credibility argument.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The mechanism is simple and powerful:

- **Attention is scarce.** Investors cannot process all available alternatives.
- **Buying and selling are asymmetric.** To buy, investors search across thousands of stocks; to sell, they typically evaluate only the few stocks they already own.
- **Attention determines the choice set; preferences determine the final choice.** A contrarian and a momentum trader may both react to the same salient stock, but choose it for different reasons.

This generates two empirical predictions:

- individual investors should be net buyers of attention-grabbing stocks;

- professional investors should display a weaker version of this pattern, or no such pattern, because they can screen more systematically and often can short.

3.2 Volume-sort design

The paper sorts stocks daily by abnormal volume and computes daily buy-sell imbalances for each investor group in each partition.

Prediction. If attention matters, then individual-investor imbalance should rise with abnormal volume. Since the top abnormal-volume decile is split into two vingtiles, the strongest evidence should appear in the highest-volume vingtile.

Interpretation. This is the cleanest proxy because unusual volume almost mechanically implies unusual investor attention, even if the source of attention is not directly observed.

3.3 Return-sort design

The paper sorts stocks by **previous-day** return and then computes next-day buy-sell imbalances.

Why lagged return?

- Investors may react only after the market closes.
- Same-day buy-sell imbalance could itself move prices.

Prediction. If extreme returns are salient, then the imbalance profile for attention-driven investors should be **U-shaped**: they should buy both yesterday’s big losers and yesterday’s big winners.

Interpretation. This is the sharpest design for separating *salience* from a simple “buy-good-news” story.

3.4 News-sort design

Stocks are partitioned into **news** and **no-news** days using the Dow Jones news feed.

Prediction. Individual investors should be more likely to buy stocks that are in the news.

Interpretation. This is the most direct attention measure, but it is also weaker empirically because some news stories are major events while others are routine press releases, and because news coverage is missing for part of the sample period.

3.5 Inference and interpretation

For each investor group and partition, the paper constructs a *time series* of daily buy-sell imbalances and studies its mean.

This is not a regression of future returns or future prices on investor activity. It is an event-style empirical design that asks: *conditional on a stock-day being salient, which investor groups tilt toward buying rather than selling?*

Interpretation. The design is deliberately low-tech and transparent. The results come from patterns in signed trading behavior, not from a highly parameterized structural model.

3.6 Interaction specification: volume, return, and news together

The paper also examines interaction effects by independently sorting:

- abnormal volume into low / medium / high bins,
- returns into low / medium / high bins,
- and then conditioning on news versus no news.

Interpretation. This design asks which attention proxy is strongest and whether multiple sources of salience reinforce one another.

3.7 Size and announcement robustness

The same sort methodology is repeated:

- separately for small, medium, and large stocks,
- and separately for earnings-announcement days, dividend-announcement days, and other days.

Interpretation. These checks target two natural concerns:

- that the results are driven only by microcaps, and
- that the results are really about broker recommendations or well-known corporate announcement windows.

3.8 Short-sale-constraint test

A competing explanation is that attention-grabbing events raise heterogeneity of beliefs, but individuals cannot easily sell short. If so, they may look like net buyers simply because the bearish side of their response is constrained.

The paper therefore focuses on trades in **stocks already owned** by investors.

- If short-sale constraints alone explain the results, attention should not materially raise the ratio of purchases to sales in already-owned stocks.
- If attention itself affects buying, then even among stocks already in the portfolio, the fraction of purchases should rise on high-attention days.

The paper also compares stocks with and without **exchange-traded options**, where short-sale constraints are looser.

3.9 Appendix model

The paper develops a simple model extending Kyle (1985) with:

- a privately informed insider,
- a market-maker,
- and **attention-driven noise traders**.

In period 2, expected noise-trader buying depends on attention:

$$E(\tilde{b}_2 \mid \tilde{y}_1) = m(A + \tilde{y}_1^2), \quad (3)$$

while expected selling depends less strongly on attention:

$$E(\tilde{s}_2 \mid \tilde{y}_1) = m(A + \kappa \tilde{y}_1^2 + (1 - \kappa)\phi^2), \quad 0 \leq \kappa < 1. \quad (4)$$

The market-maker conjectures a linear price rule,

$$P_t = \mu + \lambda d_t, \quad (5)$$

and the insider chooses a linear demand rule,

$$x_t = \alpha + \beta \tilde{y}_t. \quad (6)$$

The model delivers one asset-pricing implication:

Proposition 1. Expected noise-trader losses increase with the attention level.

Interpretation. Higher attention makes insider trading harder to detect, so attention-driven buyers should not expect superior performance. This is conceptually important because it links the trading evidence to the paper’s closing point that “all that glitters is not gold.”

4 Identification and empirical design (what makes the evidence credible)

4.1 What identifies the core evidence

The paper’s key empirical object is a **buy-sell imbalance within investor group and attention partition**. The identifying variation comes from comparing:

- low- versus high-attention stock-days,
- buys versus sells within the same investor group,
- and individuals versus professionals.

Interpretation. This makes the paper’s result much stronger than simply showing more trading

on salient days. More trading is mechanical; *net buying* is the behavioral fact.

4.2 Why “net buying” is stronger than “high trading”

Whenever abnormal volume is high, aggregate buying and aggregate selling must both be high. So attention proxies automatically pick out high-trading states.

But the paper’s central finding is that for individual investors, the composition of trading changes: buys rise relative to sells. That is the nontrivial part.

Interpretation. The paper is not rediscovering that salient stocks are traded more. It is showing that salience affects which side of the market individuals take.

4.3 Why the lagged-return design matters

The return sort uses **previous-day** returns rather than same-day returns.

This addresses two concerns:

- same-day buy pressure may mechanically move prices;
- many investors only react after observing the prior day’s performance.

Interpretation. The U-shape in next-day imbalance is therefore harder to interpret as a pure contemporaneous price-pressure artifact.

4.4 Imperfect attention proxies and measurement error

The paper is transparent that none of its proxies is perfect.

- **News** mixes major stories with routine releases.
- **Abnormal volume** can reflect block trades between institutions.
- **Extreme returns** can reflect liquidity shocks as well as salient information.

Interpretation. This imperfection mainly works against the paper. Noisy attention proxies should attenuate, not create, the observed individual-investor buying patterns.

4.5 Why the institutional comparison matters

Institutions are not just an extra sample; they are a disciplined falsification device.

- If high-attention days were simply days on which stocks were objectively attractive, professionals should also look like generalized net buyers.
- Instead, institutions trade in patterns that reflect strategy: momentum managers buy winners and sell losers; value managers do the opposite.

Interpretation. The contrast helps show that the retail pattern is not just a mechanical response to information.

4.6 Why the large-cap evidence matters

The paper shows that attention-driven buying is as strong for large-cap stocks as for small-cap stocks.

This matters because high volume in large-cap stocks is much less likely to be produced by a few idiosyncratic large trades. It weakens the claim that the volume proxy is just picking up institutional block trading or illiquidity in tiny stocks.

Interpretation. The psychology appears broad, not confined to microcaps.

4.7 Why announcement-day robustness matters

The paper shows similar patterns during:

- earnings-announcement windows,
- dividend-announcement windows,
- and other days.

Interpretation. This reduces the concern that the results are driven only by broker contact or specific corporate events with obvious directional implications.

4.8 Short-sale constraints as an alternative explanation

The paper takes the short-sale story seriously:

- salience could raise dispersion in beliefs,
- bullish investors can buy,
- bearish retail investors may be unable to short.

The crucial rebuttal is that the paper finds similar relative shifts toward buying in **stocks already owned** and also similar patterns for stocks **with and without exchange-traded options**.

Interpretation. Short-sale constraints likely contribute, but they do not fully explain the findings.

4.9 What the paper can and cannot distinguish

What the paper *can* establish:

- individual investors are more likely to buy rather than sell salient stocks;
- professionals are much less likely to show this generalized pattern;

- the result is not confined to tiny firms or to pure no-short-selling situations.

What it *cannot* fully distinguish:

- whether attention works mainly through media salience, conversations, or the stock's own price move;
- whether buyers are naive extrapolators, selective communicators, or simply boundedly rational searchers;
- whether institutions are “correct” traders in a welfare sense.

Interpretation. The paper identifies a robust behavioral pattern, not a fully structural decomposition of attention's subchannels.

4.10 Relation to the literature

The paper sits at the intersection of several literatures:

- **Disposition effect and individual trading:** investors sell winners and hold losers.
- **Visibility and trading:** volume and news increase attention.
- **Familiarity and ownership:** familiar firms attract investors.

Its distinctive contribution is to focus on **daily buy-sell imbalance** as the behavioral signature of attention. Relative to Merton's idea that investors only trade stocks they actively follow, the paper argues that attention still matters because many actual purchases seem to be drawn from the set of recently salient stocks.

5 Tables and figures

5.1 Table 1 and Figures 1a and 2a: abnormal trading volume

Read.

- For the large discount brokerage, number imbalance moves from **-18.15%** in the lowest-volume decile to **+29.50%** in the highest-volume vingtile.
- For the same sample, value imbalance moves from **-16.28%** to **+17.67%**.
- The large retail brokerage and the small discount brokerage show the same qualitative pattern.
- Institutional managers do not. Value managers, in particular, are strong net buyers of **low-volume** stocks: **+34.57%** in the lowest decile versus only **+4.83%** in the highest-volume vingtile.
- Figure 1a (simulation) and Figure 2a (data) are both upward sloping.

Economic message. High attention, as proxied by abnormal volume, attracts *retail buying*. Professionals do not respond in the same generic way.

Table 1 Buy-sell imbalances by investor type for stocks sorted on the current day's abnormal trading volume												
Decile	Large discount brokerage		Large retail brokerage		Small discount brokerage		Momentum managers		Value managers		Diversified managers	
	Number	Value	Number	Value	Number	Value	Number	Value	Number	Value	Number	Value
1 (lowest volume)	-18.15 (0.98)	-18.28 (1.37)	-25.26 (2.11)	-21.26 (1.60)	-20.49 (1.41)	-22.70 (3.88)	14.68 (1.76)	13.74 (2.26)	34.57 (5.54)	33.99 (6.45)	12.52 (2.42)	17.10 (2.91)
2	-8.90 (0.65)	-11.32 (0.98)	-18.78 (1.23)	-20.63 (1.36)	-10.31 (2.30)	-11.02 (2.47)	12.13 (1.07)	11.69 (1.44)	15.20 (2.55)	13.63 (2.91)	14.87 (1.62)	15.06 (1.97)
3	-6.23 (0.52)	-9.49 (0.84)	-15.16 (1.18)	-19.59 (1.47)	-6.95 (1.90)	-7.76 (1.47)	11.38 (0.85)	10.35 (1.15)	10.95 (1.69)	8.43 (1.93)	15.83 (1.28)	11.84 (1.65)
4	-2.76 (0.45)	-8.70 (0.73)	-10.11 (0.99)	-10.07 (1.29)	-4.92 (1.17)	-5.91 (1.56)	12.19 (0.81)	11.89 (1.07)	10.02 (1.23)	4.37 (1.61)	14.92 (1.09)	8.23 (1.50)
5	-0.76 (0.42)	-7.24 (0.67)	-4.82 (1.03)	-17.38 (1.37)	-4.06 (0.77)	-6.80 (1.34)	12.82 (0.72)	12.24 (0.94)	10.90 (1.10)	6.51 (1.38)	13.41 (0.96)	3.97 (1.28)
6	1.65 (0.42)	-7.33 (0.64)	0.23 (1.03)	-16.25 (1.17)	-1.86 (0.81)	-3.33 (1.05)	13.54 (0.70)	13.55 (0.92)	8.75 (1.03)	0.31 (1.32)	12.58 (0.90)	3.31 (1.23)
7	5.45 (0.43)	-2.87 (0.63)	4.69 (1.03)	-13.80 (1.19)	-0.05 (0.74)	-2.58 (0.96)	12.47 (0.65)	15.17 (0.85)	7.25 (0.97)	-0.61 (1.28)	10.99 (0.82)	16.61 (1.11)
8	9.20 (0.41)	-1.10 (0.62)	13.53 (1.14)	-7.92 (1.16)	1.43 (0.79)	-2.11 (0.86)	11.60 (0.64)	12.11 (0.87)	8.95 (0.95)	1.30 (1.25)	10.80 (0.84)	-4.19 (1.21)
9	13.62 (0.43)	2.86 (0.62)	19.82 (1.27)	2.02 (1.21)	5.78 (0.82)	1.36 (0.91)	11.33 (0.82)	8.90 (0.93)	7.83 (1.03)	1.09 (1.40)	11.11 (0.89)	3.47 (1.32)
10a	17.72 (0.51)	6.97 (0.78)	22.25 (1.46)	2.62 (1.24)	8.90 (0.81)	3.67 (1.07)	10.84 (0.81)	7.57 (1.22)	7.72 (1.46)	6.38 (2.04)	11.04 (1.28)	5.58 (1.93)
10b (highest volume)	29.50 (0.49)	17.67 (0.73)	19.34 (1.71)	2.02 (1.84)	17.31 (0.98)	11.78 (1.07)	6.72 (0.82)	-0.35 (1.34)	4.83 (1.79)	4.15 (2.44)	8.12 (1.37)	7.23 (2.22)

Stocks are sorted daily into deciles on the basis of the current day's abnormal volume. The decile of the highest abnormal volume is split into two vignettes (10a and 10b). Abnormal volume is calculated as the ratio of the current day's volume to the CRSP daily stock return file for NYSE, ASE, and NASDAQ stocks divided by the average volume over the previous 252 trading days. Buy-sell imbalances are reported for the trades of six groups of investors, investors at a large discount brokerage (January 1991 through November 1996), investors at a large retail brokerage (January 1997 through June 1999), investors at a small discount brokerage (January 1996 through 15 June 1999), and institutional money managers (January 1993 through March 1996) classified by the Plexus Group as following momentum, value, and diversified strategies. For each day/partition/investor group, we calculate number imbalance as number of purchases minus number of sales divided by total number of trades. Value imbalance is calculated as the value of purchases minus the value of sales divided by the total value of trades. The table reports the mean for each time series of daily imbalances for a particular investor group and partition. Standard errors, calculated using a Newey-West correction for serial dependence, appear in parentheses.

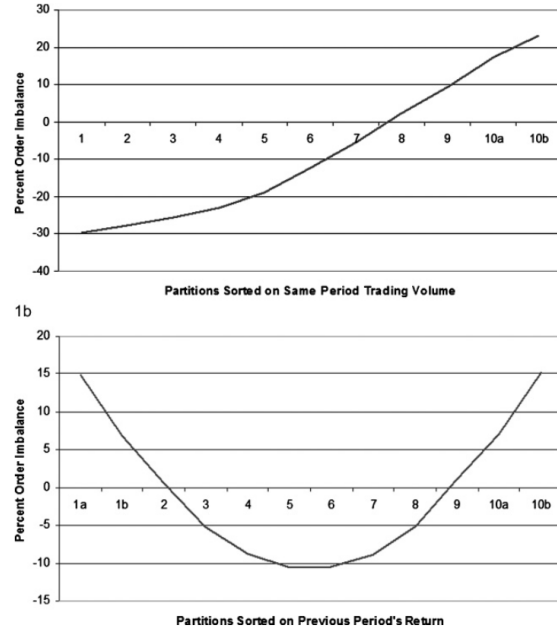


Figure 1
Simulated Buy-Sell Imbalances

5.2 Table 2 and Figures 1b and 2b: previous-day returns

Table 2 Buy-sell imbalances by investor type for stocks sorted on the previous day's return												
Decile	Large discount brokerage		Large retail brokerage		Small discount brokerage		Momentum managers		Value managers		Diversified managers	
	Number	Value	Number	Value	Number	Value	Number	Value	Number	Value	Number	Value
1a (Negative Return)	29.38 (0.61)	29.07 (0.87)	25.79 (1.60)	22.89 (1.43)	17.32 (1.04)	14.90 (1.43)	-21.03 (1.32)	-30.45 (1.83)	17.26 (3.13)	20.09 (3.41)	10.91 (2.43)	18.08 (2.88)
1b	19.19 (0.54)	16.19 (0.82)	17.66 (1.43)	1.46 (1.57)	1.12 (1.04)	8.58 (1.46)	-6.43 (1.05)	-19.21 (1.56)	14.03 (2.33)	15.62 (2.72)	15.11 (1.75)	15.11 (2.37)
2	13.69 (0.42)	8.83 (0.64)	13.73 (1.17)	5.47 (1.00)	8.65 (0.74)	3.51 (1.20)	-0.62 (0.73)	-14.58 (1.04)	11.19 (1.27)	14.18 (1.73)	10.47 (1.04)	10.47 (2.33)
3	8.86 (0.45)	3.11 (0.63)	6.60 (1.18)	-5.01 (1.09)	3.77 (0.76)	1.23 (1.23)	5.10 (0.71)	-7.72 (0.96)	10.23 (1.06)	7.68 (1.44)	12.30 (0.92)	4.75 (1.29)
4	3.94 (0.45)	-3.28 (0.64)	1.72 (1.06)	-10.98 (1.07)	1.69 (0.84)	-2.75 (1.21)	8.91 (0.76)	4.64 (1.09)	7.98 (0.99)	2.22 (1.34)	11.68 (0.96)	3.04 (1.26)
5	4.09 (0.41)	-3.57 (0.61)	-4.37 (0.95)	-14.56 (0.88)	-0.6 (0.89)	-3.68 (1.40)	9.84 (0.86)	7.02 (1.24)	9.20 (1.29)	3.69 (1.74)	11.56 (1.11)	2.62 (1.63)
6	1.82 (0.42)	-4.82 (0.62)	2.21 (0.95)	-15.85 (0.88)	-1.53 (0.89)	-4.0 (1.40)	19.31 (0.86)	25.22 (1.24)	7.92 (1.29)	0.96 (1.74)	14.00 (1.11)	1.10 (1.63)
7	2.02 (0.44)	-1.02 (0.64)	-0.07 (0.91)	-15.23 (0.91)	-1.77 (1.12)	-3.29 (0.82)	15.56 (1.28)	16.36 (0.75)	10.61 (0.99)	1.77 (1.18)	15.38 (1.55)	4.18 (0.96)
8	1.82 (0.42)	-0.62 (0.64)	0.84 (0.91)	-15.85 (0.88)	-1.53 (0.89)	-4.0 (1.40)	19.31 (0.86)	25.22 (1.24)	7.92 (1.29)	0.96 (1.74)	14.00 (1.11)	1.10 (1.63)
9	6.67 (0.43)	-4.83 (0.62)	6.54 (0.88)	-12.80 (0.88)	0.55 (1.08)	-0.79 (0.73)	22.69 (1.13)	32.44 (0.69)	4.30 (0.93)	-6.06 (1.21)	12.99 (1.66)	-1.70 (1.02)
10a	13.41 (0.51)	3.23 (0.78)	6.58 (0.90)	-11.24 (1.17)	1.17 (0.96)	-2.93 (1.41)	24.04 (1.03)	34.75 (1.37)	-4.16 (2.14)	-12.66 (2.57)	10.23 (1.58)	-3.98 (2.24)
10b (Positive Return)	23.98 (0.52)	11.13 (0.81)	9.01 (0.91)	-7.93 (1.11)	3.80 (0.84)	-3.59 (1.20)	21.50 (1.28)	36.37 (1.74)	-17.32 (3.14)	-16.83 (3.41)	7.57 (2.30)	-0.60 (2.81)

Stocks are sorted daily into deciles on the basis of the previous day's return as reported in the CRSP daily stock return file for NYSE, ASE, and NASDAQ stocks. The deciles of highest and lowest returns are each split into two vignettes (1a, 1b, 10a, and 10b). Buy-sell imbalances are reported for the trades of six groups of investors, investors at a large discount brokerage (January 1991 through November 1996), investors at a large retail brokerage (January 1997 through June 1999), investors at a small discount brokerage (January 1996 through 15 June 1999), and institutional money managers (January 1993 through March 1996) classified by the Plexus Group as following momentum, value, and diversified strategies. For each day/partition/investor group, we calculate number imbalance as number of purchases minus number of sales divided by total number of trades. Value imbalance is calculated as the value of purchases minus the value of sales divided by the total value of trades. The table reports the mean for each time series of daily imbalances for a particular investor group and partition. Standard errors, calculated using a Newey-West correction for serial dependence, appear in parentheses.

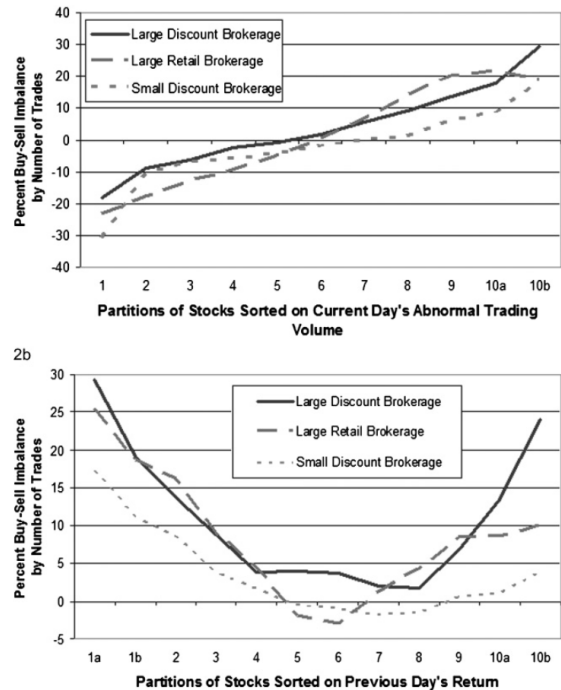


Figure 2
Individual Investor Buy-Sell Imbalances by Number of Trades for Stocks Sorted on the Current Day's Abnormal Trading Volume and Previous Day's Return

Read.

- The large discount brokerage has a clear U-shape:

- **+29.38%** number imbalance in the worst-return vingtile,
- around zero to slightly positive in middle deciles,
- **+23.98%** in the best-return vingtile.
- The value-imbalance version is even sharper on the negative side: **+29.07%** for the worst-return vingtile, falling to **-8.62%** in return decile 8, then rebounding to **+11.13%** for the best-return vingtile.
- The large retail brokerage also shows a U-shape, but with a weaker response to big winners.
- The small discount brokerage mostly buys yesterday's losers, not yesterday's winners.
- Institutional managers behave according to style:
 - momentum managers: **-21.03%** in the worst-return vingtile and **+21.50%** in the best-return vingtile,
 - value managers: **+17.26%** in the worst-return vingtile and **-17.32%** in the best-return vingtile.

Economic message. Individuals react to salience in a broad, non-strategy-specific way; professionals react in a way that matches their investment styles.

5.3 Table 3: news sorts

Read.

- For all days, the large discount brokerage has number imbalance **9.35%** on news days versus **2.70%** on no-news days.
- For the large retail brokerage, the same comparison is **16.17%** versus **-1.84%**.
- Conditional on positive return days, the large discount brokerage has imbalance **1.74%** with news versus **-2.51%** without news.
- Conditional on negative return days, the same comparison is **17.39%** with news versus **8.86%** without news.

Economic message. News coverage shifts individual investors toward buying, and it especially amplifies the effect of salient negative-return events.

5.4 Figure 3: volume, return, and news together

Read.

- Buy-sell imbalances rise with abnormal volume in each return bin.
- At the large discount brokerage, low-return and high-return stocks tend to have larger imbalances than middle-return stocks within a volume bin.

Table 3
Buy-Sell Imbalances by Investor Type for Stocks Sorted on the Current Day's News

	Large discount brokerage		Large retail brokerage		Small discount brokerage		Momentum managers		Value managers		Diversified managers	
Partition	Number imbalance	Value imbalance	Number imbalance	Value imbalance	Number imbalance	Value imbalance	Number imbalance	Value imbalance	Number imbalance	Value imbalance	Number imbalance	Value imbalance
Panel A: All Days												
News	9.35 (0.72)	0.07 (0.86)	16.17 (1.29)	-2.36 (1.32)	6.76 (0.48)	1.97 (0.72)	13.38 (1.33)	14.00 (1.71)	6.36 (1.59)	-0.24 (2.05)	6.21 (1.11)	2.26 (1.50)
No News	2.70 (0.43)	-5.62 (0.63)	-1.84 (0.87)	-14.59 (0.97)	-0.66 (0.58)	-4.87 (1.23)	12.20 (1.11)	10.43 (1.60)	0.96 (1.37)	3.62 (1.49)	7.26 (0.97)	1.24 (0.84)
Panel B: Positive Return Days												
News	1.74 (0.94)	-9.25 (1.07)	14.07 (1.04)	-7.74 (1.25)	1.14 (0.64)	-3.13 (0.95)	22.70 (1.50)	31.85 (2.10)	5.87 (1.94)	-1.01 (2.65)	7.80 (1.31)	3.92 (2.00)
No News	-2.51 (0.54)	-14.31 (0.79)	1.76 (0.88)	-13.90 (1.00)	-4.49 (1.00)	-8.41 (0.79)	22.29 (1.31)	25.64 (1.46)	16.20 (1.51)	6.67 (1.74)	8.95 (1.05)	4.66 (1.05)
Panel C: Negative Return Days												
News	17.39 (0.83)	10.91 (1.12)	15.59 (1.58)	3.17 (1.43)	13.77 (0.71)	9.32 (1.08)	3.94 (1.43)	-7.39 (2.11)	4.29 (2.09)	-2.41 (2.77)	4.72 (1.30)	2.24 (2.25)
No News	8.86 (0.53)	3.85 (0.81)	-3.38 (0.88)	-13.57 (0.85)	4.35 (0.77)	1.29 (1.42)	0.68 (1.25)	-8.60 (1.46)	6.92 (1.52)	1.60 (1.89)	5.58 (1.03)	-4.11 (1.23)
Panel D: Zero Return Days												
News	1.41 (1.76)	-5.90 (2.31)	-0.44 (0.94)	-8.74 (1.45)	1.58 (2.25)	-1.22 (2.68)	14.12 (2.35)	15.16 (3.19)	11.37 (3.44)	9.59 (4.35)	5.21 (2.47)	1.62 (3.68)
No News	-0.95 (0.68)	-6.40 (1.13)	-14.49 (1.00)	-18.24 (1.08)	-3.27 (1.35)	-7.95 (2.04)	14.60 (1.38)	12.86 (1.81)	10.65 (1.73)	2.42 (2.49)	8.36 (1.27)	-0.17 (1.84)

Stocks are partitioned daily into those with and without news stories (reported by the Dow Jones News Service) that day. On average, there is no news for 91% of stocks. Buy-sell imbalances are reported for the trades of six groups of investors, investors at a large discount brokerage (January 1991 through November 1996), investors at a large retail brokerage (January 1997 through June 1999), investors at a small discount brokerage (January 1996 through 15 June 1999), and institutional money managers (January 1993 through March 1996) classified by the Plexus Group as following momentum, value, and diversified strategies. Buy-sell imbalances are reported for all stocks and days with or without news. They are also reported separately for the days on which stocks had positive, negative, and zero returns. For each day/partition/investor group, we calculate number imbalance as number of purchases minus number of sales divided by total number of trades. Value imbalance is calculated as the value of purchases minus the value of sales divided by the total value of trades. The table reports the mean for each time series of daily imbalances for a particular investor group and partition. Standard errors, calculated using a Newey-West correction for serial dependence, appear in parentheses.

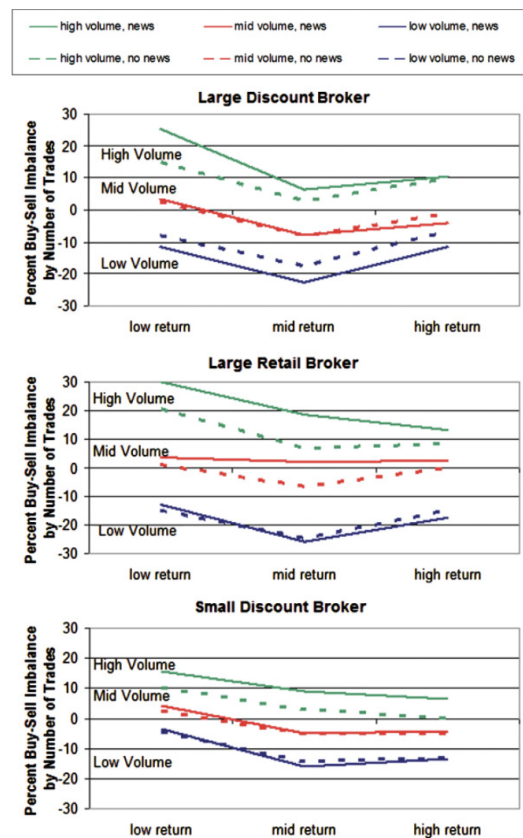


Figure 3
Buy-Sell Imbalances for Investors at a Large Discount Brokerage Based on 3-by-3 Partition on Abnormal Volume and Returns, Conditional on News Coverage

- The paper concludes from this exercise that **abnormal volume is the single best attention proxy**, returns come second, and the simple news indicator is the least informative.

Economic message. The three salience measures reinforce each other, but volume carries the strongest signal of attention-driven buying.

5.5 Table 4 and Figure 4: size and announcement robustness

Table 4
Buy-Sell Imbalances for Large Discount Brokerage Investors for Stocks Sorted on the Current Day's Abnormal Trading Volume, the Previous Day's Return, and the Current Day's News and Then Partitioned on Market Capitalization

Decile	Small Stocks		Mid Cap Stocks		Large Stocks	
	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance
1 (lowest volume)	-16.11 (1.17)	-13.35 (1.50)	-18.43 (2.36)	-17.18 (2.49)	-31.89 (6.32)	-30.33 (6.46)
2	-5.04 (0.86)	-4.37 (1.18)	-12.09 (1.19)	-14.16 (1.50)	-21.44 (2.32)	-22.17 (2.49)
3	-2.23 (0.72)	-2.49 (1.04)	-6.66 (0.85)	-9.24 (1.19)	-15.81 (1.29)	-15.35 (1.56)
4	3.22 (0.71)	0.16 (1.01)	-1.99 (0.67)	-6.65 (1.01)	-9.17 (0.76)	-12.01 (1.11)
5	6.22 (0.70)	2.98 (1.01)	1.34 (0.67)	-4.30 (1.01)	-5.46 (0.58)	-9.99 (0.87)
6	9.44 (0.65)	5.74 (0.96)	2.94 (0.62)	-5.00 (0.85)	-1.24 (0.54)	-9.12 (0.77)
7	10.90 (0.64)	4.47 (0.97)	6.03 (0.59)	-0.99 (0.92)	-3.27 (0.54)	-0.76 (0.76)
8	11.83 (0.61)	5.42 (0.92)	6.80 (0.57)	-1.88 (0.89)	9.38 (0.56)	-0.80 (0.77)
9	15.13 (0.53)	7.27 (0.83)	9.27 (0.59)	-0.98 (0.85)	14.50 (0.64)	4.54 (0.84)
10a	16.94 (0.64)	7.73 (0.99)	12.97 (0.76)	3.80 (1.05)	19.76 (1.22)	11.13 (0.99)
10b (highest volume)	20.77 (0.54)	32.13 (0.83)	24.41 (0.86)	15.04 (1.12)	28.26 (1.33)	21.65 (1.53)

In panel A, stocks are sorted daily into deciles on the basis on the current day's abnormal volume. The decile of highest abnormal volume is split into two vingtiles (10a and 10b). Abnormal volume is calculated as the ratio of the current day's volume (as reported in the CRSP daily stock return files for NYSE, ASE, and NASDAQ stocks) divided by the average volume over the previous 252 trading days. In Panel B, stocks are sorted daily into deciles on the basis of the previous day's return as reported in the CRSP daily stock return files for NYSE, ASE, and NASDAQ stocks. The deciles of highest and lowest returns are each split into two vingtiles (1a, 1b, 10a and 10b). Abnormal trading volume is calculated as the ratio of the current day's trading volume (as reported in the CRSP daily stock return files for NYSE, ASE, and NASDAQ stocks) divided by the average trading volume over the previous 252 trading days. In Panel C, stocks are partitioned daily into those with and without news stories that day (as reported by the Dow Jones News Service). On average there is no news for 91 percent of stocks. For all three panels, after sorting and partitioning, stocks are further separated into three groups based on market capitalization. We use monthly New York Stock Exchange market equity breakpoints to form our size groups. Each month we classify all stocks (both NYSE-listed and non-listed stocks) with market capitalization less than or equal to the thirtieth percentile breakpoint as small stocks, stocks with market capitalization greater than thirtieth percentile and less than or equal to the seventieth percentile as medium stocks, and stocks with market capitalization greater than the seventieth percentile as large stocks. Buy-sell imbalances are reported for the trades of investors at a large discount brokerage (January 1991 through November 1996). For each day/partition/size group, we calculate number imbalance as number of purchases minus number of sales divided by total number of trades. Value imbalance is calculated as the value of purchases minus the value of sales divided by the total value of trades. The table reports the mean for each time series of daily imbalances for a particular investor group and partition. Standard errors, calculated using a Newey-West correction for serial dependence, appear in parentheses.

Table 4
Panel B: Buy-Sell Imbalances for Stocks Sorted First on the Previous Day's Return and Then on Market Capitalization

Decile	Small Stocks		Mid Cap Stocks		Large Stocks	
	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance
1a (Negative Return)	24.88 (0.66)	26.06 (0.99)	32.71 (1.25)	30.83 (1.48)	38.73 (1.92)	34.55 (2.15)
1b	14.37 (0.65)	12.61 (0.99)	17.61 (0.96)	14.99 (1.27)	25.26 (1.38)	21.93 (1.62)
2	10.69 (0.54)	6.30 (0.82)	9.67 (0.86)	4.99 (0.89)	18.53 (0.67)	13.50 (0.92)
3	6.97 (0.65)	2.05 (0.96)	5.06 (0.59)	-0.95 (0.86)	11.09 (0.59)	5.35 (0.82)
4	4.48 (0.53)	-3.23 (0.78)	0.87 (0.62)	-5.29 (0.90)	4.23 (0.60)	-3.06 (0.81)
5	3.72 (0.42)	-3.64 (0.63)	3.59 (0.46)	-4.45 (0.69)	4.02 (0.47)	-3.58 (0.67)
6	0.42 (0.42)	0.62 (0.62)	0.49 (0.49)	0.73 (0.73)	0.54 (0.54)	0.75 (0.75)
7	5.28 (0.54)	-2.63 (0.79)	2.87 (0.60)	-4.84 (0.59)	0.80 (0.59)	-8.23 (0.81)
8	8.88 (0.61)	2.78 (0.93)	2.07 (0.56)	-7.78 (0.85)	-0.83 (0.58)	-10.96 (0.80)
9	11.98 (0.54)	5.49 (0.83)	6.73 (0.61)	-5.41 (0.90)	3.31 (0.67)	-6.69 (0.90)
10a	16.88 (0.63)	10.59 (0.96)	12.09 (0.82)	2.53 (1.14)	5.53 (1.25)	-1.81 (1.48)
10b (Positive Return)	26.98 (0.57)	18.69 (0.88)	20.83 (1.06)	8.19 (1.33)	7.76 (1.84)	2.94 (2.06)

Table 4
Panel C: Buy-Sell Imbalances for Stocks Sorted First on Market Capitalization and Then on Current Day's News

Decile	Small Stocks		Mid Cap Stocks		Large Stocks	
	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance
News	19.87 (1.47)	14.59 (1.85)	13.38 (1.15)	3.87 (1.62)	6.52 (0.85)	-1.35 (0.97)
All Days	7.53 (0.48)	2.82 (0.70)	3.12 (0.57)	-4.83 (0.88)	-2.91 (0.67)	-9.86 (0.94)

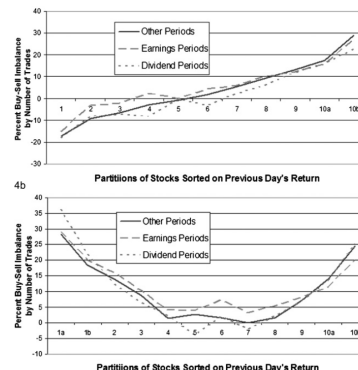


Figure 4
Buy-Sell Imbalances for Investors at a Large Discount Brokerage During Earnings Announcement Periods, Dividend Announcement Periods, and Other Periods for Volume Deciles (top) and Return Deciles (bottom)

Read.

- The abnormal-volume pattern is present for all size groups. For **large-cap stocks**, number imbalance goes from **-31.89%** in the lowest-volume decile to **+28.26%** in the highest-volume vingtile.
- The return pattern is also present across size groups. For **large-cap stocks**, imbalance is **+38.73%** in the worst-return vingtile and remains positive at **+7.76%** in the best-return vingtile.
- On news days, even large-cap stocks show stronger buying: **+6.52%** with news versus **-2.91%** without news.
- Figure 4 shows that the volume and return patterns are qualitatively similar during earnings-announcement periods, dividend-announcement periods, and other periods.

Economic message. The results are not a microcap curiosity and are not driven only by a narrow set of standard corporate announcement events.

5.6 Table 5: stocks already owned

Read.

Table 5
Buy-sell imbalances for large discount brokerage investors for stocks already owned by each investor.
stocks sorted on the current day's abnormal trading volume, the previous day's return, and the current
day's news
Panel A: Buy-Sell Imbalances for Stocks Already Owned Sorted on Current Day's Abnormal Trading
Volume

Decile	Large Discount Brokerage		Large Retail Brokerage		Small Discount Brokerage	
	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance	Number Imbalance	Value Imbalance
1 (lowest volume)	-54.22 (1.43)	-55.64 (1.89)	-28.74 (1.42)	-33.99 (1.84)	-24.25 (6.28)	-33.22 (7.58)
2	-51.13 (0.78)	-53.20 (1.07)	-29.46 (1.09)	-34.09 (1.36)	-33.80 (3.18)	-29.67 (4.47)
3	-48.27 (0.64)	-49.69 (0.95)	-29.34 (1.04)	-31.25 (1.31)	-31.76 (1.71)	-30.05 (2.44)
4	-47.19 (0.56)	-49.51 (0.88)	-28.69 (0.94)	-32.96 (1.11)	-35.65 (1.26)	-33.93 (1.96)
5	-45.95 (0.53)	-47.59 (0.81)	-26.71 (0.90)	-31.04 (1.07)	-32.34 (1.12)	-30.01 (1.63)
6	-45.01 (0.49)	-48.65 (0.71)	-24.32 (0.90)	-29.71 (1.04)	-30.00 (0.97)	-26.50 (1.42)
7	-42.36 (0.50)	-45.85 (0.71)	-21.83 (0.84)	-30.29 (0.89)	-29.85 (0.95)	-26.21 (1.33)
8	-39.43 (0.51)	-43.75 (0.71)	-18.72 (0.81)	-27.21 (0.87)	-28.20 (0.87)	-26.23 (1.22)
9	-35.64 (0.52)	-40.68 (0.70)	-15.45 (0.78)	-21.79 (0.91)	-27.07 (0.85)	-24.99 (1.21)
10a	-33.03 (0.63)	-39.31 (0.85)	-12.27 (0.97)	-19.97 (1.12)	-26.81 (1.06)	-27.99 (1.42)
10b (highest volume)	-24.97 (0.69)	-32.82 (0.92)	-15.01 (1.04)	-20.04 (1.19)	-17.32 (0.98)	-19.38 (1.42)

In panel A, stocks are sorted daily into deciles on the basis of the current day's abnormal volume. The decile of the highest abnormal volume is split into two vingtiles (10a and 10b). Abnormal volume is calculated as the ratio of the current day's volume (as reported in the CRSP daily stock return files for NYSE, ASE, and NASDAQ stocks) divided by the average volume over the previous 252 trading days. In panel B, stocks are sorted daily into deciles on the basis of the previous day's return as reported in the CRSP daily stock return files for NYSE, ASE, and NASDAQ stocks. The deciles of highest and lowest returns are each split into two vingtiles (1a, 1b, 10a, and 10b). Abnormal trading volume is calculated as the ratio of the current day's trading volume (as reported in the CRSP daily stock return files for NYSE, ASE, and NASDAQ stocks) divided by the average trading volume over the previous 252 trading days. In panel C, stocks are partitioned daily into those with and without news stories that day (as reported by the Dow Jones News Service). Buy-sell imbalances are reported for the trades of investors at a large discount brokerage (January 1991 through November 1996), investors at a large retail brokerage (January 1997 through June 1999), and investors at a small discount brokerage (January 1996 through December 1998). Imbalances are calculated for purchases and sales by investors of stocks already held by each investor's account. For each day/partition/investor group, we calculate number imbalance as number of purchases minus number of sales divided by total number of trades. Value imbalance is calculated as the value of purchases minus the value of sales divided by the total value of trades. The table reports the mean for each time series of daily imbalances for a particular investor group and partition. Standard errors, calculated using a Newey-West correction for serial dependence, appear in parentheses.

- Because investors usually sell stocks they already own, all imbalances in this table are negative on average.
- The key fact is the *relative* change with attention. For the large discount brokerage, number imbalance for already-owned stocks rises from **-54.22%** in the lowest-volume decile to **-24.97%** in the highest-volume vingtile.
- For news sorts, the large discount brokerage has **-40.91%** on news days versus **-45.05%** on no-news days.
- For return sorts, the large discount brokerage becomes much less negative after large losses: **-9.68%** in the worst-return vingtile versus about **-48%** in middle-to-high deciles.

Economic message. Even when short selling is irrelevant because the stock is already owned, attention shifts behavior toward *buying more*. This is the paper's cleanest rebuttal to the pure short-sale-constraint story.

6 Short summary

- Individual investors are net buyers of salient stocks, measured by high abnormal volume, extreme prior returns, and news coverage.
- Professionals do not show this generalized pattern; their trading looks much more strategy-

specific.

- The result survives checks by market capitalization, announcement windows, and already-owned stocks.
- Attention appears to shape the buying choice set and may help explain why retail investors buy stocks that later disappoint.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a clear behavioral fact: **individual investors buy attention-grabbing stocks**. They do so much more than professionals, and they do so in a way that is strongest for stocks with unusual trading volume and for stocks with extreme recent returns.

7.2 Economic intuition

The intuition is not that individuals lack preferences or strategies. Rather, when there are thousands of possible purchases, *attention chooses which stocks get considered at all*. Selling is different because the investor usually evaluates only a small existing portfolio. That asymmetry makes attention much more important for buys than for sells.

7.3 Takeaway

The paper provides a persuasive microfoundation for attention-driven trading in retail markets. What becomes salient gets bought. Because salience need not line up with value, the result helps explain why retail trading can amplify sentiment and why active stock picking by households often disappoints.